Module 4: Mathematical Logic

Propositional logic, equivalences, normal forms,
predicates, quantifiers and rules of inference

Exam Focus

Scope. Syllabus 4.1 to 4.3: mathematical logic (propositional and predicate logic, propositional equivalences, normal forms, predicates and quantifiers, nested quantifiers, rules of inference).
Logic appeared in four of the five papers in the set, always with an OR pair, and it is the module where the answer is a *method* rather than a calculation. Learn the truth-table method and the algebraic method equally well, because both are asked.

Contents

1 Propositions and connectives 2

2 Tautology, contradiction, contingency 2

3 Logical equivalences 2
3.1 Converse, inverse, contrapositive . . . 3

4 Proving equivalences 3
4.1 Truth-table method . . . 3
4.2 Algebraic method (without a truth table) . . . 4

5 Predicates and quantifiers 5
5.1 Symbolizing statements . . . 5
5.2 Nested quantifiers . . . 6

6 Rules of inference 6

7 Normal forms 7

8 Answer skeletons 8

9 Self-test 9

10 Last-minute page 9

Propositions and connectives

A **proposition** is a declarative statement with exactly one truth value: true (T) or false (F). Questions, commands, and open sentences containing an unspecified variable are not propositions.

Connective	Notation	Meaning	Truth condition
Negation	$\neg p$	not p	True exactly when p is false
Conjunction	$p \wedge q$	p and q	True when both are true
Inclusive OR	$p \vee q$	p or q or both	False only when both are false
Exclusive OR	$p \oplus q$	exactly one of p, q	True when their values differ
Implication	$p \to q$	if p, then q	False only for $p = T, q = F$
Biconditional	$p \leftrightarrow q$	p iff q	True when their values agree

For n variables, a complete truth table has 2^n rows. Note the two key rewrites:

$$p \to q \equiv \neg p \vee q, \qquad p \leftrightarrow q \equiv (p \to q) \wedge (q \to p).$$

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \to q$	$p \leftrightarrow q$
T	T	F	T	T	F	T	T
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

Tautology, contradiction, contingency

- A **tautology** is true on every row, e.g. $p \vee \neg p$.
- A **contradiction** is false on every row, e.g. $p \wedge \neg p$.
- A **contingency** is true on some rows and false on others, e.g. $p \wedge q$.

To classify a formula, construct its full truth table and inspect the final column.

Past-paper question

Oct 2024 CAE-2 Q2A(a). Check whether the statement $(p \vee \neg p) \vee \big((q \vee r) \vee \neg(q \vee r)\big)$ is a tautology or a contradiction.

Solution

Simplify first. $(p \vee \neg p)$ is a tautology, so the whole formula is true regardless of the other part. Equally, $(q \vee r) \vee \neg(q \vee r)$ is a tautology. A disjunction of tautologies is a tautology.
Full truth table (8 rows for p, q, r):

p	q	r	$p \vee \neg p$	$q \vee r$	$\neg(q \vee r)$	$(q \vee r) \vee \neg(q \vee r)$	Full formula
T	T	T	T	T	F	T	T
T	T	F	T	T	F	T	T
T	F	T	T	T	F	T	T
T	F	F	T	F	T	T	T
F	T	T	T	T	F	T	T
F	T	F	T	T	F	T	T
F	F	T	T	T	F	T	T
F	F	F	T	F	T	T	T

The final column is T on every row, so the formula is a **tautology**.

Logical equivalences

Two formulas are logically equivalent, written $P \equiv Q$, when their truth values agree for every assignment.

Law	Form 1	Form 2
Identity	$p \wedge T$	p; $\quad p \vee F = p$
Domination	$p \vee T$	T; $\quad p \wedge F = F$
Idempotent	$p \vee p$	p; $\quad p \wedge p = p$
Double negation	$\neg\neg p$	p
Commutative	$p \vee q$	$q \vee p$; similarly for $\wedge$
Associative	$(p \vee q) \vee r$	$p \vee (q \vee r)$; similarly for $\wedge$
Distributive	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
De Morgan	$\neg(p \wedge q)$	$\neg p \vee \neg q$
De Morgan	$\neg(p \vee q)$	$\neg p \wedge \neg q$
Absorption	$p \vee (p \wedge q)$	p; dual: $p \wedge (p \vee q) = p$
Implication	$p \to q$	$\neg p \vee q$
Biconditional	$p \leftrightarrow q$	$(p \to q) \wedge (q \to p)$

Do Not Confuse

The distributive law also works in the other direction: $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$. Students often remember only one version.

Converse, inverse, contrapositive

For $p \to q$: the **converse** is $q \to p$; the **inverse** is $\neg p \to \neg q$; the **contrapositive** is $\neg q \to \neg p$.

The original and the contrapositive are equivalent. The converse and the inverse are equivalent to each other, but are not generally equivalent to the original.

Past-paper question

Nov 2024 Section A Q8. Write the converse and the contrapositive of: "If a function is differentiable then it is continuous."

Solution

Original: $p \to q$ where p = "the function is differentiable", q = "the function is continuous".
Converse: $q \to p$, that is, "If a function is continuous then it is differentiable." (False in general.)
Contrapositive: $\neg q \to \neg p$, that is, "If a function is not continuous then it is not differentiable." (True, since it is equivalent to the original.)
The inverse would be: "If a function is not differentiable then it is not continuous."

Do Not Confuse

For "if p, then q," do not swap the hypothesis and conclusion and assume the result is equivalent. Only the contrapositive is guaranteed equivalent.

Proving equivalences

Two methods are asked for, sometimes in the same paper as an OR pair.

Truth-table method

Build a truth table with a column for every intermediate sub-expression. If the two sides produce identical columns, they are equivalent.

Past-paper question

Nov 2024 Q16(a)(i). Using the truth table, prove the logical equivalence $\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (q \wedge \neg p)$.

Solution

p	q	$p \leftrightarrow q$	$p \wedge \neg q$	$q \wedge \neg p$	$(p \wedge \neg q) \vee (q \wedge \neg p)$	$\neg(p \leftrightarrow q)$
T	T	T	F	F	F	F
T	F	F	F	T	T	T
F	T	F	T	F	T	T
F	F	T	T	T	T	F

The final two columns are identical, so $\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (q \wedge \neg p)$.

Past-paper question

Nov 2024 Q16(b)(ii). Using the truth table, prove the logical equivalence $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.

Solution

p	q	r	$q \vee r$	$p \wedge (q \vee r)$	$p \wedge q$	$p \wedge r$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T
T	F	T	T	T	F	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

The columns match, proving the distributive law. Note this is exactly the *contradiction* test too: the formula is false only in the row $p = T, q = F, r = F$, where it is a contingency.

Algebraic method (without a truth table)

Start from one side, apply one named identity per step, and reach the other side.

Past-paper question

Nov 2024 Q16(b)(i). Without using a truth table, show that $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$.

Solution

$$
\begin{aligned}
p \leftrightarrow q &\equiv (p \to q) \wedge (q \to p) && \text{biconditional} \\
&\equiv (\neg p \vee q) \wedge (\neg q \vee p) && \text{implication} \\
&\equiv \big((\neg p \wedge \neg q) \vee (\neg p \wedge p)\big) \vee \big((q \wedge \neg q) \vee (q \wedge p)\big) && \text{distributive} \\
&\equiv (\neg p \wedge \neg q) \vee F \vee F \vee (q \wedge p) && \text{complement} \\
&\equiv (\neg p \wedge \neg q) \vee (p \wedge q) && \text{identity} \\
&\equiv (p \wedge q) \vee (\neg p \wedge \neg q) && \text{commutative}
\end{aligned}
$$

Past-paper question

Aug 2025 Q3B(b). Without using a truth table, prove that $\big[(p \vee q) \wedge \neg p\big] \to \neg q \equiv q \to p$.

Solution

Work with the left side:

$$\begin{aligned}
\big[(p \vee q) \wedge \neg p\big] \to \neg q &\equiv \neg\big[(p \vee q) \wedge \neg p\big] \vee \neg q && \text{implication}\\
&\equiv \neg(p \vee q) \vee \neg\neg p \vee \neg q && \text{De Morgan, double negation}\\
&\equiv (\neg p \wedge \neg q) \vee p \vee \neg q && \text{De Morgan}\\
&\equiv p \vee (\neg p \wedge \neg q) \vee \neg q && \text{associative, commutative}\\
&\equiv p \vee \neg q && \text{absorption: } p \vee (\neg p \wedge \neg q) = p\\
&\equiv q \to p && \text{implication}
\end{aligned}$$

Exam Focus

Worked, tautology without a truth table. Show $[(p \vee q) \wedge \neg p] \to q$ is a tautology.

$$\begin{aligned}
[(p \vee q) \wedge \neg p] \to q &\equiv \neg[(p \vee q) \wedge \neg p] \vee q\\
&\equiv \neg(p \vee q) \vee p \vee q\\
&\equiv (\neg p \wedge \neg q) \vee p \vee q\\
&\equiv p \vee q\\
&\equiv T.
\end{aligned}$$

Since the formula is equivalent to T, it is a tautology.

Predicates and quantifiers

A **predicate** $P(x)$ is a statement whose truth depends on the value assigned to x. A domain specifies the allowed values.

- $\forall x\, P(x)$: for every x in the domain, $P(x)$ is true.
- $\exists x\, P(x)$: there is at least one x for which $P(x)$ is true.

Remember

Negation rules: $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$ and $\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$. For example, "not everyone passed" means "someone did not pass," not "nobody passed."

Symbolizing statements

Statement	Symbolic form
All men are giants	$\forall x(M(x) \to G(x))$
Some men are giants	$\exists x(M(x) \wedge G(x))$
No man is a giant	$\forall x(M(x) \to \neg G(x))$
Not every man is a giant	$\exists x(M(x) \wedge \neg G(x))$
All men are giants and all women are giants	$\forall x M(x) \to G(x)$ *and* $\forall x W(x) \to G(x)$

Past-paper question

Oct 2025 Q4(a)(iii). If p means "he is intelligent" and q means "he is strong", write the symbolic form of: "It is wrong that he is intelligent or strong."

Solution

"It is wrong that" is a negation. So the statement is $\neg(p \vee q)$, which by De Morgan is $\neg p \wedge \neg q$, that is, "he is neither intelligent nor strong."

Past-paper question

Nov 2024 Section A Q5. Symbolize the statement "All men are giants".

Solution

Let $M(x)$ be "x is a man" and $G(x)$ be "x is a giant".

$$\forall x\,(M(x) \to G(x)).$$

Note the implication, not a conjunction: a universal statement saying all men are giants does not assert that every object is both a man and a giant.

Nested quantifiers

$\forall x \exists y\, R(x, y)$ says each x may have its own y; $\exists y \forall x\, R(x, y)$ says one fixed y works for every x. In general these are not equivalent.

To disprove a universal statement, give a counterexample; to prove an existential statement, give a witness.

Past-paper question

Oct 2025 Q4(b)(iii). Represent the following in predicate calculus: (a) Some dragon is sleeping. (b) No dragon is sleeping. (c) Every dragon is sleeping. (d) Not every dragon is sleeping.

Solution

Let $D(x)$ be "x is a dragon" and $S(x)$ be "x is sleeping".
(a) $\exists x\,(D(x) \wedge S(x))$.
(b) $\neg\exists x\,(D(x) \wedge S(x))$, equivalently $\forall x\,(D(x) \to \neg S(x))$.
(c) $\forall x\,(D(x) \to S(x))$.
(d) $\neg\forall x\,(D(x) \to S(x))$, equivalently $\exists x\,(D(x) \wedge \neg S(x))$.

Rules of inference

An argument is valid when its conclusion follows from its premises.

Rule	Premises	Conclusion
Modus ponens	$p \to q, p$	q
Modus tollens	$p \to q, \neg q$	$\neg p$
Hypothetical syllogism	$p \to q, q \to r$	$p \to r$
Disjunctive syllogism	$p \vee q, \neg p$	q
Addition	p	$p \vee q$
Simplification	$p \wedge q$	p (or q)
Conjunction	p, q	$p \wedge q$
Resolution	$p \vee q, \neg p \vee r$	$q \vee r$

Past-paper question

Nov 2024 Q16(a)(ii). Explain the inference rules modus ponens and modus tollens.

Solution

Modus ponens: from $p \to q$ and p, infer q. In words: if p implies q and p is true, then q is true.
Modus tollens: from $p \to q$ and $\neg q$, infer $\neg p$. In words: if p implies q and q is false, then p must be false.

Past-paper question

Aug 2025 Q4A. Show that the following set of premises is inconsistent: $P \to Q$, $P \to R$, $Q \to \neg R$, and P.

Solution

Assume P. From $P \to Q$ and modus ponens, Q. From $P \to R$ and modus ponens, R. From $Q \to \neg R$ and modus ponens, $\neg R$. But R and $\neg R$ cannot both be true.
Since the premises lead to a contradiction, the set is **inconsistent**.
Equivalently, from $P \to R$ we get R, and from $Q \to \neg R$ we get $Q \to \neg R$, which by modus tollens with $\neg R$ gives $\neg Q$. Combined with Q this is also a contradiction.

Past-paper question

Oct 2025 Q4(a)(ii). Show that $\neg P$ is a valid conclusion from $P \to Q$ and $\neg Q$.

Solution

This is modus tollens. Premise 1: $P \to Q$. Premise 2: $\neg Q$. Conclusion: $\neg P$.

	Statement	Reason
1	$P \to Q$	Premise
2	$\neg Q$	Premise
3	$\neg P$	1, 2, modus tollens

Past-paper question

Oct 2025 Q4(b)(i). If A works hard, then either B or C will enjoy themselves. If B enjoys himself, then A will not work hard. If D enjoys himself, then C will not. Therefore, if A works hard, D will not enjoy himself. Show that these statements constitute a valid argument.

Solution

Let A = A works hard, B = B enjoys, C = C enjoys, D = D enjoys.
Premises: $A \to (B \vee C)$; $B \to \neg A$; $D \to \neg C$. Conclusion: $A \to \neg D$.

	Statement	Reason
1	$A \to (B \vee C)$	Premise
2	$B \to \neg A$	Premise
3	$D \to \neg C$	Premise
4	A	Assumption (conditional proof)
5	$B \vee C$	1, 4, modus ponens
6	$\neg B$	2 and 4, modus tollens
7	C	5, 6, disjunctive syllogism
8	$\neg C$	3 and 7, modus tollens
9	$\neg D$	3 and 8, modus tollens
10	$A \to \neg D$	4–9, conditional proof

Since line 10 follows from the premises, the argument is valid.

Normal forms

An **elementary product** (minterm) is an AND of literals; an **elementary sum** (maxterm) is an OR of literals. A literal is a variable or its negation.

- **DNF**: OR of AND terms, e.g. $(p \wedge q) \vee (\neg p \wedge r)$.
- **CNF**: AND of OR terms, e.g. $(p \vee q) \wedge (\neg p \vee r)$.

To convert algebraically: remove $\to$ and $\leftrightarrow$, push negations inward using De Morgan's laws, then distribute. For canonical forms use the truth table: DNF takes the rows where the formula is true (minterms); CNF takes the rows where it is false (maxterms).

Past-paper question

Oct 2025 Q4(a)(i). Obtain the principal disjunctive normal forms of $\neg P \vee Q$.

Solution

Build the truth table for $F = \neg P \vee Q$ over P, Q:

P	Q	$\neg P$	Q	F=$\neg P \vee Q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	F	T

F is false only in the row $P = T, Q = F$. The principal DNF lists the minterm for every row where F is true, with each variable appearing in its true (or negated) form:

$$F = (P \wedge Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q).$$

Exam Focus

Worked, PDNF and CNF of $p \to q$. The formula is false only when $p = T, q = F$.

$$\text{PDNF} = (p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q),$$

$$\text{PCNF} = (\neg p \vee q).$$

Past-paper question

Oct 2025 Q4(a)(iv) and Nov 2024 Section A Q11. What is conjunctive normal form?

Solution

Conjunctive normal form (CNF) is a formula written as a conjunction (AND) of clauses, where each clause is a disjunction (OR) of literals. The canonical form, the principal conjunctive normal form, contains one maxterm for every row where the formula is false.
Example: $p \to (q \wedge r)$ has CNF $(\neg p \vee q) \wedge (\neg p \vee r)$.

Answer skeletons

Answer Skeleton

Truth table (3 to 9 marks). List the variables; make all 2^n assignments; add a column for each intermediate connective; evaluate the target column; classify as tautology, contradiction or contingency and state the result.

Answer Skeleton

Equivalence proof (5 to 9 marks). Write the expression; remove implications; apply the named equivalence law one step at a time; reach the target expression. Alternatively, show identical truth-table columns. Name each law to make the reasoning clear.

Answer Skeleton

Inference or validity (5 to 9 marks). Number the premises; identify the rule applied at each line; show the conclusion as the last line. For a conditional conclusion, state the assumption and use conditional proof.

Answer Skeleton

Quantifier negation (3 marks). Negate the quantifier, switch $\forall \leftrightarrow \exists$, then negate the predicate.

Answer Skeleton

Normal form (3 to 9 marks). Write the truth table, identify the rows where the formula is true (for DNF) or false (for CNF), then write one minterm or maxterm per row with all variables present.

Self-test

1. Define proposition, predicate, tautology, contradiction and contingency.
2. State the truth condition for implication and biconditional.
3. State De Morgan's laws and the negation rules for quantifiers.
4. Define DNF and CNF, with one example of each.
5. Find the converse, inverse and contrapositive of "If a number is divisible by 4, then it is even."
6. Negate: "Every student solved some problem." Express the result with quantifiers.
7. Identify the inference rule: $p \to q$, $q \to r$, therefore $p \to r$.
8. Prove $p \to (q \to p)$ is a tautology without a truth table.
9. Prove by truth table that $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
10. Obtain the principal disjunctive normal form of $\neg p \vee q$.
11. Show that the premises $p \to q$, $q \to r$, $r \to \neg p$ are inconsistent.
12. Represent in predicate calculus: every student passed at least one exam.

Answers

1. Proposition: a declarative statement with one truth value. Predicate: a statement whose truth depends on an unspecified variable. Tautology: true on every row. Contradiction: false on every row. Contingency: true on some rows and false on others.
2. $p \to q$ is false only when $p = T, q = F$; $p \leftrightarrow q$ is true exactly when p and q agree.
3. $\neg(p \wedge q) \equiv \neg p \vee \neg q$; $\neg(p \vee q) \equiv \neg p \wedge \neg q$; $\neg \forall x P \equiv \exists x \neg P$; $\neg \exists x P \equiv \forall x \neg P$.
4. DNF is an OR of AND terms, e.g. $(p \wedge q) \vee (\neg p \wedge r)$. CNF is an AND of OR terms, e.g. $(p \vee q) \wedge (\neg p \vee r)$.
5. Converse: "If a number is even then it is divisible by 4." Inverse: "If a number is not divisible by 4 then it is not even." Contrapositive: "If a number is not even then it is not divisible by 4." The original and the contrapositive are equivalent.
6. $\neg\big[\forall x\, \exists y\, S(x,y)\big] \equiv \exists x\, \forall y\, \neg S(x,y)$: there is a student who solved no exam.
7. Hypothetical syllogism.
8. $p \to (q \to p) \equiv \neg p \vee (\neg q \vee p) \equiv p \vee \neg p \vee \neg q \equiv T$.
9. Eight-row truth table; the columns for $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$ are identical.
10. $(p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$.
11. From $p \to q$ and $q \to r$ and $r \to \neg p$, assume p to get q, then r, then $\neg p$, a contradiction. Hence inconsistent.
12. $\forall x\, \exists y\, S(x,y)$.

Last-minute page

CONNECTIVES
$p \to q \equiv \neg p \vee q$
$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
n variables $\Rightarrow 2^n$ rows

LAWS
$\neg(p \wedge q) \equiv \neg p \vee \neg q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
$p \vee (p \wedge q) \equiv p$

QUANTIFIERS
$\neg \forall x P \equiv \exists x \neg P$
$\neg \exists x P \equiv \forall x \neg P$
$\forall x \exists y R \neq \exists y \forall x R$

INFERENCE
Modus ponens: $p \to q$, $p \Rightarrow q$
Modus tollens: $p \to q$, $\neg q \Rightarrow \neg p$
Hyp. syllogism, disj. syllogism, resolution

NORMAL FORMS
DNF: OR of AND terms
CNF: AND of OR terms
PDNF: minterms of the TRUE rows
PCNF: maxterms of the FALSE rows

DO NOT CONFUSE
Only the contrapositive matches the original.
$\forall$ vs $\exists$ changes meaning on negation.
P alone is not a proposition.

Name the law at each step. "Transforming" without naming the law earns half marks in a 9-mark proof.